Data Handling: Import, Cleaning and Visualisation

Exercise/Workshop 6: Visualisation

Prof. Dr. Ulrich Matter

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Exercises

Exercise A: Summary Stats and Data Display

Install and load the R package `nycflights13` and load `tidyverse`.

```
# install and load packages
# install.packages("nycflights13")
library(nycflights13)
library(tidyverse)
library(haven)
```

Loading the `nycflights13`-package automatically imports a large `data.frame/tibble` called `flights`. Have a look at the dimensions, data types, variables, and first few observations. Also use `str()` to get a summary overview over the structure of the dataset.

```
flights
```

```
## # A tibble: 336,776 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time arr_delay carrier
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>     <dbl> <chr>
##  1  2013     1     1      517            515         2      830            819        11 UA
##  2  2013     1     1      533            529         4      850            830        20 UA
##  3  2013     1     1      542            540         2      923            850        33 AA
##  4  2013     1     1      544            545        -1     1004           1022       -18 B6
##  5  2013     1     1      554            600        -6      812            837       -25 DL
##  6  2013     1     1      554            558        -4      740            728        12 UA
##  7  2013     1     1      555            600        -5      913            854        19 B6
##  8  2013     1     1      557            600        -3      709            723       -14 EV
##  9  2013     1     1      557            600        -3      838            846        -8 B6
## 10  2013     1     1      558            600        -2      753            745         8 AA
## # ... with 336,766 more rows, and 9 more variables: flight <int>, tailnum <chr>, origin <chr>,
## #   dest <chr>, air_time <dbl>, distance <dbl>, hour <dbl>, minute <dbl>, time_hour <dttm>
```

```
str(flights)
```

```
## tibble [336,776 x 19] (S3: tbl_df/tbl/data.frame)
##  $ year          : int [1:336776] 2013 2013 2013 2013 2013 2013 2013 2013 2013 2013 ...
##  $ month         : int [1:336776] 1 1 1 1 1 1 1 1 1 1 ...
##  $ day           : int [1:336776] 1 1 1 1 1 1 1 1 1 1 ...
##  $ dep_time      : int [1:336776] 517 533 542 544 554 554 555 557 557 558 ...
##  $ sched_dep_time: int [1:336776] 515 529 540 545 600 558 600 600 600 600 ...
##  $ dep_delay     : num [1:336776] 2 4 2 -1 -6 -4 -5 -3 -3 -2 ...
```

```
##  $ arr_time      : int [1:336776] 830 850 923 1004 812 740 913 709 838 753 ...
##  $ sched_arr_time: int [1:336776] 819 830 850 1022 837 728 854 723 846 745 ...
##  $ arr_delay     : num [1:336776] 11 20 33 -18 -25 12 19 -14 -8 8 ...
##  $ carrier       : chr [1:336776] "UA" "UA" "AA" "B6" ...
##  $ flight        : int [1:336776] 1545 1714 1141 725 461 1696 507 5708 79 301 ...
##  $ tailnum       : chr [1:336776] "N14228" "N24211" "N619AA" "N804JB" ...
##  $ origin        : chr [1:336776] "EWR" "LGA" "JFK" "JFK" ...
##  $ dest          : chr [1:336776] "IAH" "IAH" "MIA" "BQN" ...
##  $ air_time      : num [1:336776] 227 227 160 183 116 150 158 53 140 138 ...
##  $ distance      : num [1:336776] 1400 1416 1089 1576 762 ...
##  $ hour          : num [1:336776] 5 5 5 5 6 5 6 6 6 6 ...
##  $ minute        : num [1:336776] 15 29 40 45 0 58 0 0 0 0 ...
##  $ time_hour     : POSIXct[1:336776], format: "2013-01-01 05:00:00" "2013-01-01 05:00:00" "2013-01-0
```

The dataset is on flights departing from New York City in 2013 (original data provided by the US Bureau of Transportation Statistics). You can have a closer look at what the variables describe by looking at the provided documentation: `?flights`.

The following table summarises and nicely displays one of the key variables in the dataset. Use your data preparation, data analysis, and data display skills to reproduce the table (output in either HTML, LaTeX/PDF, or Word format).

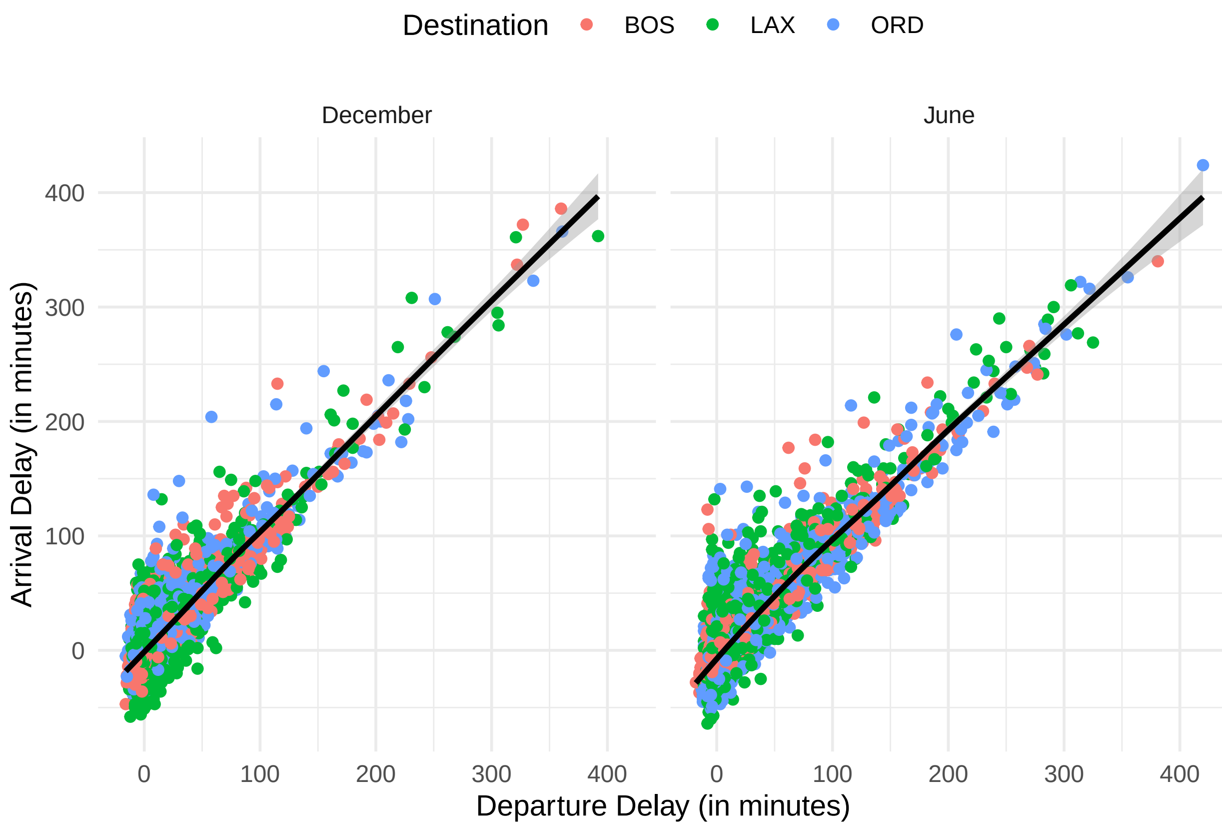
Table 1: Summary Statistics: Air time (in minutes) for flights between NYC and BOS (2013)

Month	Mean	Median	Std. Deviation	Min. Value	Max. Value
January	39.05	39	4.99	23	81
February	38.72	38	4.62	30	86
March	39.57	39	5.80	21	82
April	38.08	37	4.93	29	68
May	39.28	39	4.72	29	82
June	38.76	38	5.20	29	112
July	39.08	38	5.96	30	99
August	39.58	39	4.28	30	66
September	39.09	39	4.00	26	70
October	38.60	38	4.50	31	63
November	38.95	39	4.19	30	62
December	38.55	38	5.61	30	73

Hint: The `lubridate` package might be helpful to parse the month column. The function `months()` (base package) extracts the months from `Date` variables.

Exercise B: R Graphics with `ggplot2`

Again working with the `flights` dataset, reproduce the following figure with `ggplot()`. Hint: Some preparations are needed (filtering, formatting, and removal of missing values). Note that the figure only compares flights in December with flights in June for the destinations "BOS", "ORD", and "LAX".



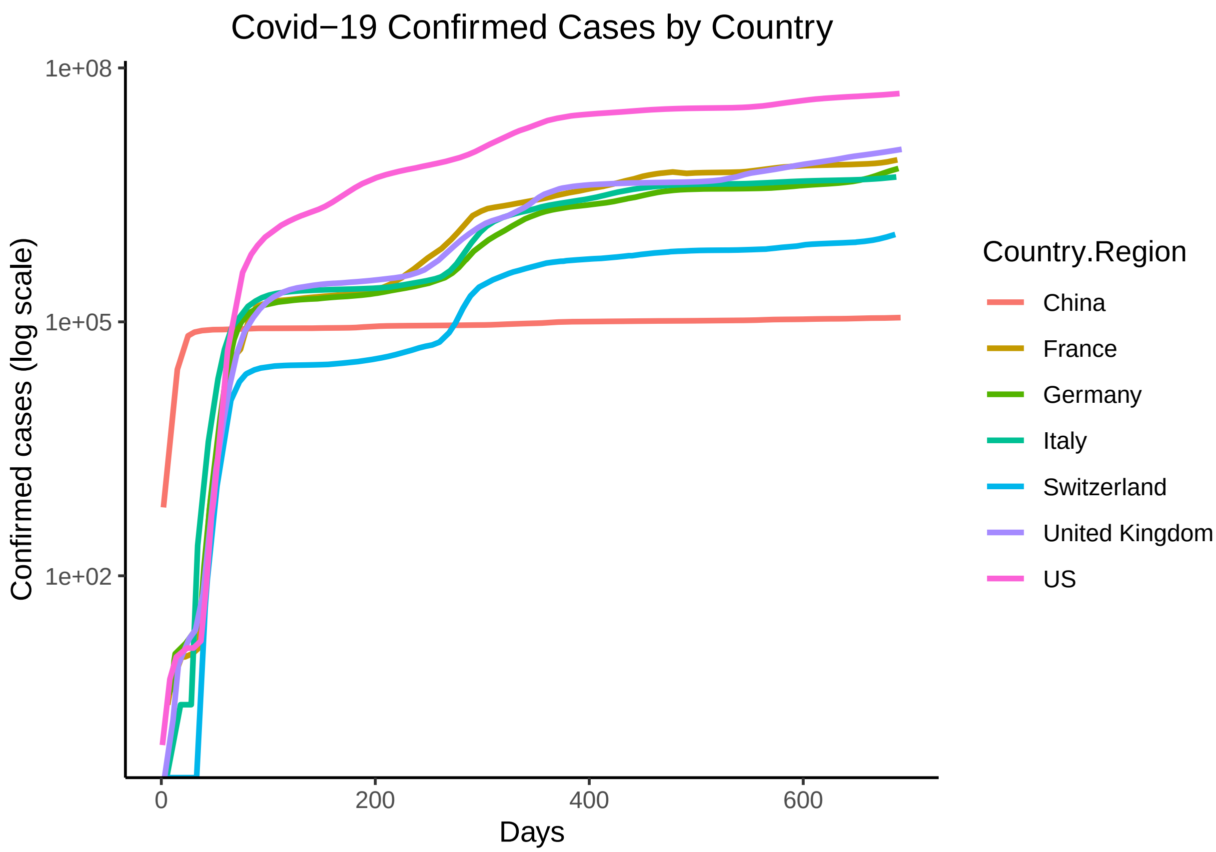
Exercise C: ggplot2 challenge on COVID

Plot the cumulative confirmed COVID cases (on a log scale) for China, France, Germany, Italy, Switzerland, the United Kingdom, and the US. Use John Hopkinds GitHub data. Your graph will look like the example shown below.

Hint: download the data as follows:

```
confirmedraw <- read.csv(
    "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/
    csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_confirmed_global.csv")
```

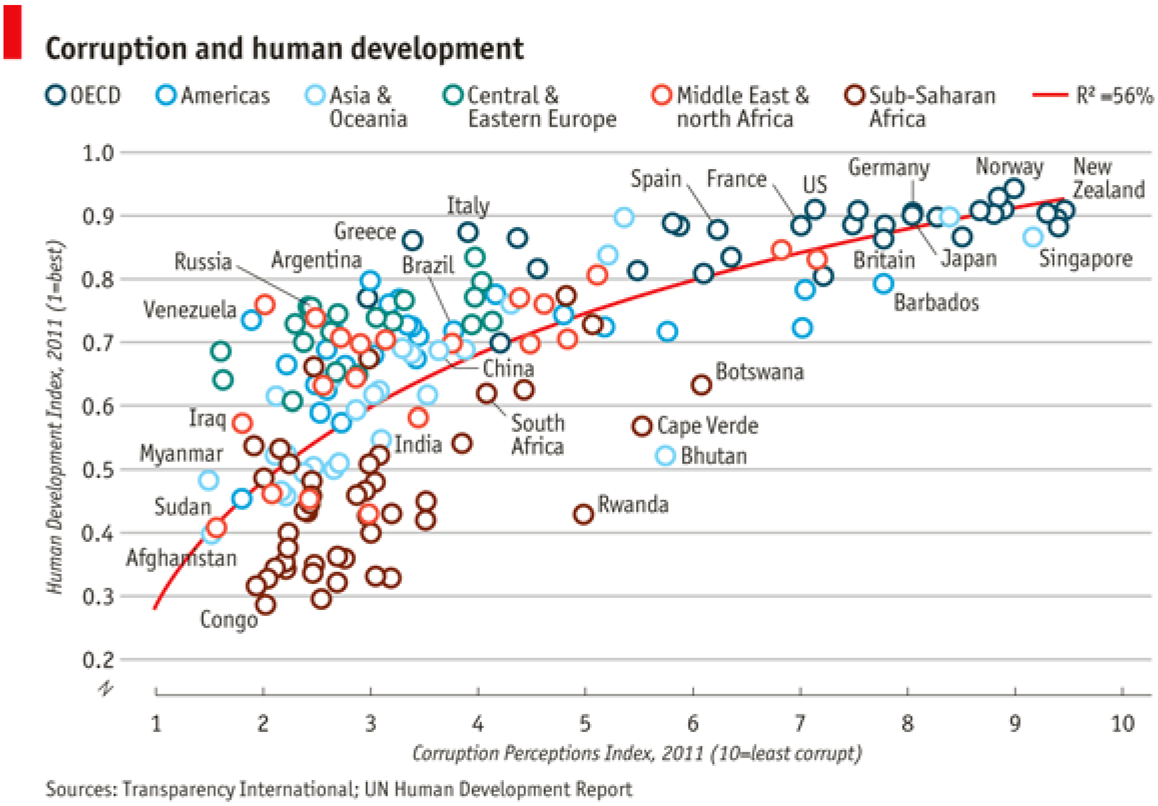
Various cleaning steps are required (gathering data, reformatting date values, calculating the cumulative sum of cases, . . .). For plotting, replace the dates by numbers (the first date is day 1, the second date day 2, etc. – as shown in the graph). This exercise is inspired by https://mdl.library.utoronto.ca/technology/tutorials/covid-19-data-r.



Additional Exercises (advanced)

Exercise D: ggplot2 challenge on corruption

Import the dataset `EconomistData.csv` (provided in `data.zip`) to R. The underlying data sources have been used by The Economist (2nd December 2011, 'Corrosive Corruption') to generate the following figure:



Given the data, the challenge is to reproduce (or get as close to) the original figure with `ggplot2`.

Exercise E: How confident are you from confidence intervals?

A new 2021 study looks at whether poor people are more in favor of redistributive policies (https://www.aeaweb.org/articles?id=10.1257/pol.20190276). The authors conduct a randomized survey experiment with over 30,000 participants across 10 countries, half of whom are informed of their position in the national income distribution. In general, they find that people who are told they are relatively poorer than they thought are less concerned about inequality and are not more supportive of redistribution.

```
crosscountry <- read_dta("D:/HSG_teaching/Data_Handling/datahandling-exercises/06_visualization/data/cr
write_csv(crosscountry, "../data/poor_redistribution.csv")
crosscountry <- crosscountry %>%
  select(context, country)

poor_redistribution <- read_csv("../data/poor_redistribution.csv")
```

```
## Rows: 48960 Columns: 19

## -- Column specification ----------------------------------------------------------------------------
## Delimiter: ","
## chr  (3): country, country_name, age
## dbl (16): dwell_time_total, context, weight, age_group, gender, q06_vote, q01_currently_distribu...
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
poor_redistribution <- na.omit(poor_redistribution)
```

Christmas Bonus Exercise (advanced)

Two PhD students, after a long evening, were arguing about fundamental research questions. Talking about drug restrictions, one of the two was convinced that the liberalization of private drug use (such as cannabis and harder narcotics) would naturally lead to an increase in violence and in episodes of (collective) psychosis. The only way to see if this claim holds water is to find data that would confirm this hypothesis. The two economists then gathered the two following datasets: - To measure psychotic episodes, they used data from the National UFO Reporting Center, which collects data on each reported UFO seeing across the world. The data are available here: http://www.nuforc.org/webreports.html. The data are ready for use in the file `ufodata_exercise.csv`, which contains the following variables: - `State.name` and `State.abbr` give the US State identifier; - `year` gives the year; - `total.views.year` gives the number of UFO seeings per year and per State - To measure drug liberalization policies, they scraped the Wikipedia website to get a timeline of cannabis legalization dates across US States. The data was obtained here: https://en.wikipedia.org/wiki/Legality_of_cannabis_by_U.S._jurisdiction. The data are ready for use in the file `pot_exercise.csv`, which contains the following variables: - `state` gives the US State identifier; - `year` gives the year; - `Legalization_medical` is 1 if the use of cannabis for medical purposes was legalized in the State-year, and 0 otherwise.

The hypothesis is the following: the legalization of cannabis for medical purposes leads to an increase in hallucinations and episodes of "psychosis". These episodes of psychosis might translate into the number of UFO seeings. Since cannabis was not legalized at the same time in all States ("staggered" introduction), we can compare States which have introduced legal cannabis for medical purposes with States which did not.

First, import the two datasets `pot_exercise.csv` and `ufodata_exercise.csv` found on Canvas. Explore the datasets, and merge the two datasets using `left_join()` into a new dataframe called `data`. Clean the new dataset.

```r
pot <- read_csv("../data/pot_exercise.csv")
```

```
## Rows: 2400 Columns: 3

## -- Column specification --------------------------------------------------------------------------------
## Delimiter: ","
## chr (1): state
## dbl (2): year, Legalization_medical

##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
ufodata <- read_csv("../data/ufodata_exercise.csv")
```

```
## Rows: 2683 Columns: 4

## -- Column specification --------------------------------------------------------------------------------
## Delimiter: ","
## chr (2): State.name, State.abbr
## dbl (2): year, total.views.year

##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Merge
data <- left_join(pot, ufodata,
                  by = c("state" = "State.name", "year"))

# Cleaning: total.views.year is NA for many observations. Remove these observations.
data <- data[!is.na(data$total.views.year), ]
```

You want to see whether the number of UFO seeings in States which have legalized cannabis for medical use is larger than in States which have not. You consider the following linear regression model:

$$Y = \alpha + \beta \text{Lega} + u,$$

where Y is the number of UFO seeings , and Lega indicates whether the State has legalized cannabis for medical use. Save your regression model in an object called `regression`.

```
regression <- lm(total.views.year ~ Legalization_medical, data = data)
summary(regression)
```

```
##
## Call:
## lm(formula = total.views.year ~ Legalization_medical, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -89.07 -45.46 -34.46   9.29 819.54
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            49.463      1.809  27.336  < 2e-16 ***
## Legalization_medical   48.602     12.767   3.807 0.000144 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 85.72 on 2288 degrees of freedom
## Multiple R-squared:  0.006294,   Adjusted R-squared:  0.00586
## F-statistic: 14.49 on 1 and 2288 DF,  p-value: 0.0001444
```

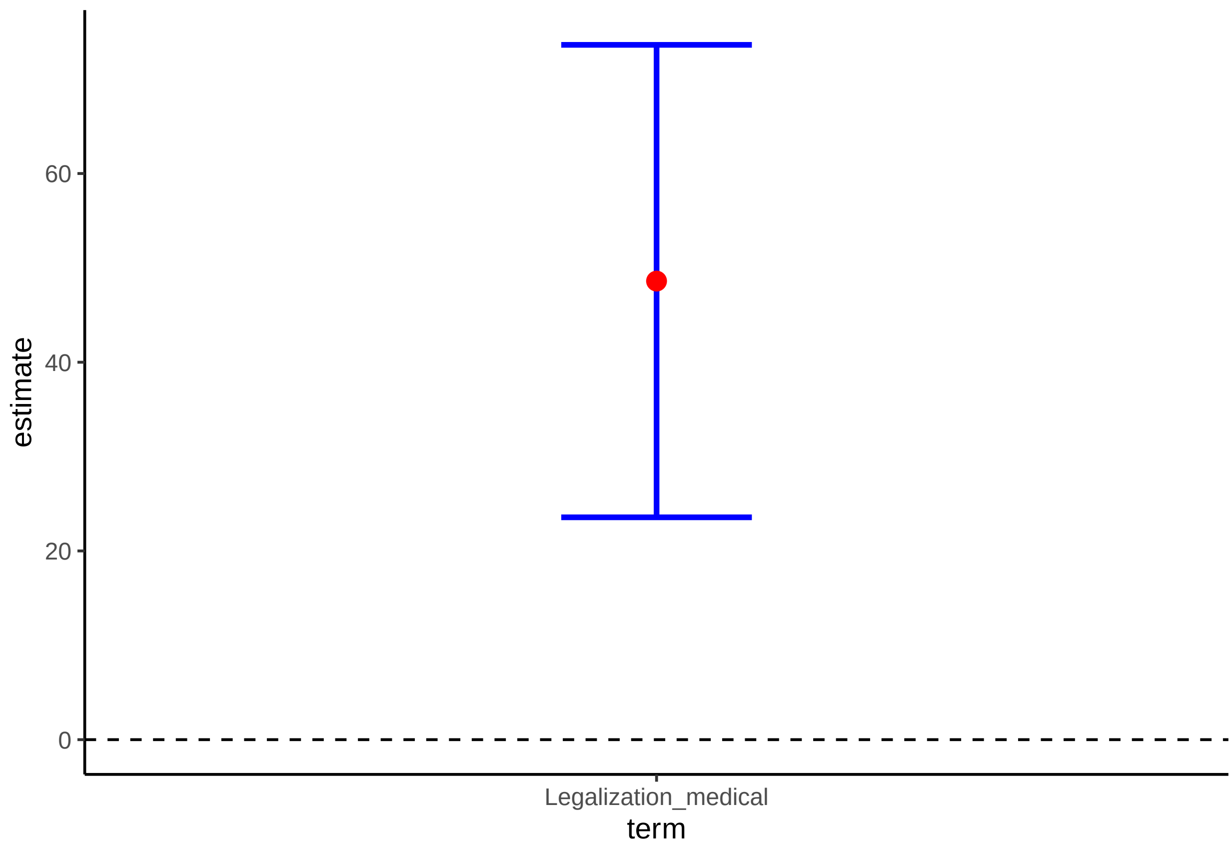
Use `ggplot()` to plot the coefficient estimates of Legalization_medical and corresponding confidence intervals. Interpret the results.

(*Hint*: To draw a plot of regression estimates, proceed as follows: (1) transform the regression output into a tibble specifying that you need 95% confidence intervals. A handy way to do this is to use the command `tidy()`from the `broom` package (load the package accordingly); (2) on your graph, you will need `geom_point()` to draw your point estimate, `geom_errorbar()` to draw the confidence intervals, and `geom_hline()` to draw the horizontal zero line. Note that you don't need to draw the coefficient for the constant.)

```
library(broom)
summary.regression <- tidy(regression, conf.int = T, conf.level = 0.95)

# We are not interested in the constant:
summary.regression <- filter(summary.regression, term == "Legalization_medical")

ggplot(summary.regression, aes(x = term, y = estimate)) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width=0.2, size=1, color="blue") +
  geom_point(size = 3, color = "red") +
  geom_hline(yintercept = 0, linetype = 2) +
  theme_classic()
```



Interpretation: having cannabis legalized increases the average number of UFO seeings by 48.60. This effect is statistically significant at the 95% level (the confidence intervals do not touch the 0-line).

In this analysis, we did not take into account the fact that States are heterogeneous, and that our effect could be driven only by the fact that States with lots of UFO seeings are also States that would be more inclined to have cannabis legalized. Perform the same analysis as before, but this time include State and year controls in your regression. This model is called a "Difference-in-Difference" model with many groups and time periods in econometrics. Look at the coefficient of `Legalization_medical`. What happened?

```r
regression_2 <- lm(total.views.year ~ Legalization_medical + as.factor(state) + as.factor(year), data =
summary(regression_2)
```

```
##
## Call:
## lm(formula = total.views.year ~ Legalization_medical + as.factor(state) +
##     as.factor(year), data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -237.78  -21.65   -0.81   21.60  452.49
##
## Coefficients:
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       -22.4804    10.8172  -2.078 0.037807 *
## Legalization_medical               -5.8074     7.8983  -0.735 0.462253
## as.factor(state)Alaska            -12.9307    10.8840  -1.188 0.234945
## as.factor(state)Arizona            69.7893    10.7131   6.514 9.03e-11 ***
## as.factor(state)Arkansas           -0.5721    10.7688  -0.053 0.957637
```

```
## as.factor(state)California     276.8101    10.7131  25.838  < 2e-16 ***
## as.factor(state)Colorado        36.0608    10.7689   3.349 0.000826 ***
## as.factor(state)Connecticut     13.3726    10.7131   1.248 0.212075
## as.factor(state)Delaware       -24.8881    11.3801  -2.187 0.028849 *
## as.factor(state)Florida        126.7893    10.7131  11.835  < 2e-16 ***
## as.factor(state)Georgia         26.6643    10.7131   2.489 0.012886 *
## as.factor(state)Hawaii         -15.1107    11.1495  -1.355 0.175466
## as.factor(state)Idaho           -0.1342    10.9511  -0.012 0.990220
## as.factor(state)Illinois        57.5184    10.7131   5.369 8.75e-08 ***
## as.factor(state)Indiana         26.2059    10.7131   2.446 0.014517 *
## as.factor(state)Iowa            -1.1654    10.8270  -0.108 0.914291
## as.factor(state)Kansas          -2.7405    10.8870  -0.252 0.801279
## as.factor(state)Kentucky         7.6497    10.7662   0.711 0.477455
## as.factor(state)Louisiana       -2.9119    10.7661  -0.270 0.786823
## as.factor(state)Maine           -2.8532    10.8843  -0.262 0.793237
## as.factor(state)Maryland        10.0539    10.7662   0.934 0.350490
## as.factor(state)Massachusetts   25.4143    10.7131   2.372 0.017766 *
## as.factor(state)Michigan        44.7893    10.7131   4.181 3.02e-05 ***
## as.factor(state)Minnesota       16.2476    10.7131   1.517 0.129510
## as.factor(state)Mississippi    -10.6210    10.8842  -0.976 0.329264
## as.factor(state)Missouri        28.8518    10.7131   2.693 0.007132 **
## as.factor(state)Montana         -6.3784    10.9496  -0.583 0.560272
## as.factor(state)Nebraska       -12.9716    10.9507  -1.185 0.236329
## as.factor(state)Nevada           8.6346    10.7689   0.802 0.422751
## as.factor(state)New Hampshire   -1.9292    10.8240  -0.178 0.858555
## as.factor(state)New Jersey      29.7268    10.7131   2.775 0.005570 **
## as.factor(state)New Mexico       7.5184    10.7131   0.702 0.482881
## as.factor(state)New York        82.3309    10.7131   7.685 2.29e-14 ***
## as.factor(state)North Carolina  43.8934    10.7131   4.097 4.33e-05 ***
## as.factor(state)North Dakota   -31.0125    11.5557  -2.684 0.007335 **
## as.factor(state)Ohio            58.4768    10.7131   5.458 5.35e-08 ***
## as.factor(state)Oklahoma         3.3161    10.7689   0.308 0.758159
## as.factor(state)Oregon          43.0601    10.7131   4.019 6.03e-05 ***
## as.factor(state)Pennsylvania    67.1851    10.7131   6.271 4.30e-10 ***
## as.factor(state)Rhode Island   -19.6230    11.3002  -1.737 0.082614 .
## as.factor(state)South Carolina  18.9769    10.7689   1.762 0.078175 .
## as.factor(state)South Dakota   -27.5703    11.4600  -2.406 0.016220 *
## as.factor(state)Tennessee       19.1276    10.8242   1.767 0.077347 .
## as.factor(state)Texas           88.1226    10.7131   8.226 3.29e-16 ***
## as.factor(state)Utah             5.2268    10.7131   0.488 0.625681
## as.factor(state)Vermont        -16.5941    11.0763  -1.498 0.134233
## as.factor(state)Virginia        26.7933    10.7144   2.501 0.012468 *
## as.factor(state)Washington     110.7476    10.7131  10.338  < 2e-16 ***
## as.factor(state)West Virginia   -8.9998    10.9436  -0.822 0.410948
## as.factor(state)Wisconsin       23.3309    10.7131   2.178 0.029528 *
## as.factor(state)Wyoming        -24.1491    11.2940  -2.138 0.032608 *
## as.factor(year)1974              1.4205    10.8281   0.131 0.895640
## as.factor(year)1975              3.3507    10.7090   0.313 0.754401
## as.factor(year)1976              4.7939    10.6013   0.452 0.651172
## as.factor(year)1977              2.4137    10.7091   0.225 0.821701
## as.factor(year)1978              3.7075    10.7614   0.345 0.730491
## as.factor(year)1979              0.9610    10.8275   0.089 0.929282
## as.factor(year)1980              0.1974    10.8246   0.018 0.985450
## as.factor(year)1981             -1.2791    10.8276  -0.118 0.905970
```

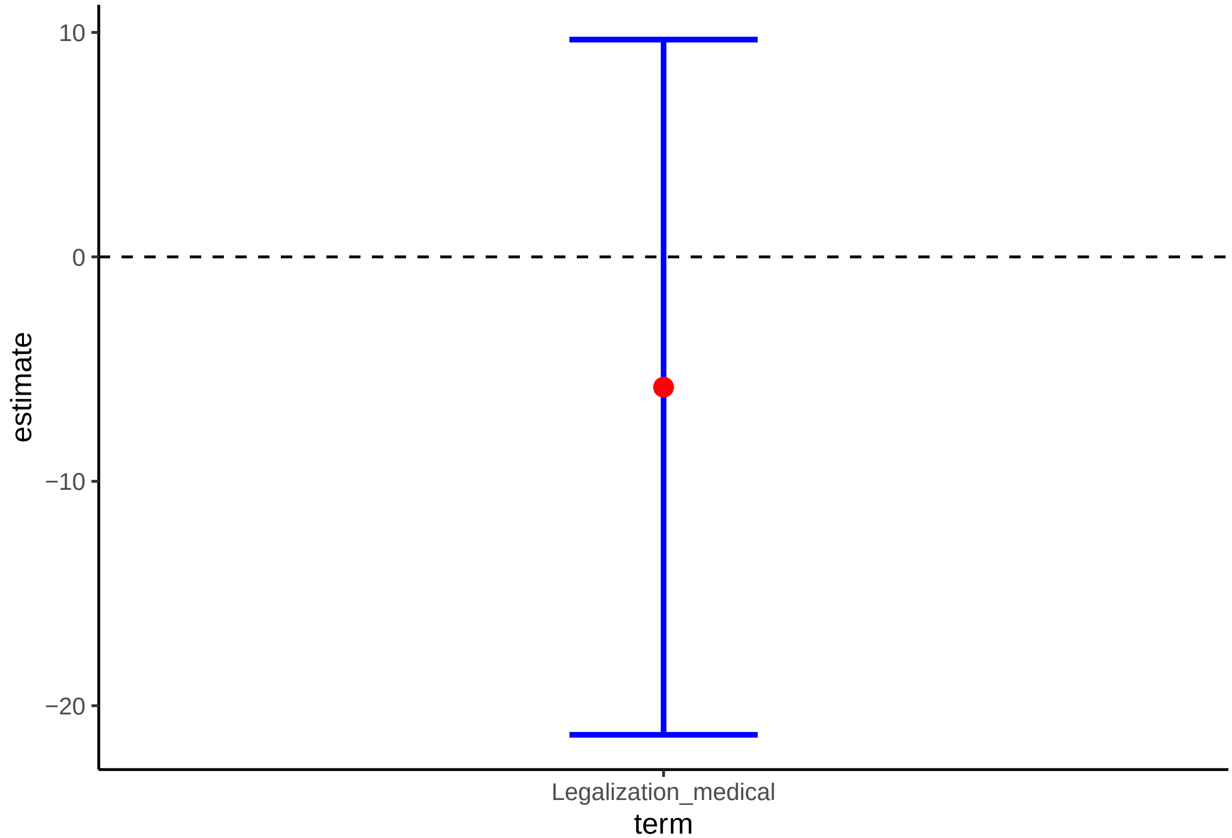
```
## as.factor(year)1982             -0.5927    10.8309  -0.055 0.956366
## as.factor(year)1983              0.1295    10.8900   0.012 0.990512
## as.factor(year)1984             -0.5451    10.8842  -0.050 0.960062
## as.factor(year)1985             -2.3502    11.0208  -0.213 0.831148
## as.factor(year)1986             -3.6177    10.9470  -0.330 0.741072
## as.factor(year)1987              0.5093    10.8244   0.047 0.962473
## as.factor(year)1988             -2.1824    11.0918  -0.197 0.844036
## as.factor(year)1989              1.9680    10.7062   0.184 0.854173
## as.factor(year)1990             -0.2347    10.8909  -0.022 0.982811
## as.factor(year)1991             -2.2650    11.0212  -0.206 0.837189
## as.factor(year)1992              0.9808    10.7610   0.091 0.927389
## as.factor(year)1993              3.8233    10.6525   0.359 0.719696
## as.factor(year)1994              5.9252    10.7057   0.553 0.580002
## as.factor(year)1995             26.8698    10.5490   2.547 0.010928 *
## as.factor(year)1996             15.7362    10.6540   1.477 0.139812
## as.factor(year)1997             24.8698    10.5490   2.358 0.018483 *
## as.factor(year)1998             35.0383    10.5597   3.318 0.000921 ***
## as.factor(year)1999             55.1860    10.5502   5.231 1.85e-07 ***
## as.factor(year)2000             53.0983    10.5597   5.028 5.35e-07 ***
## as.factor(year)2001             58.5298    10.5490   5.548 3.23e-08 ***
## as.factor(year)2002             58.6898    10.5490   5.564 2.97e-08 ***
## as.factor(year)2003             69.7698    10.5490   6.614 4.69e-11 ***
## as.factor(year)2004             76.8460    10.5502   7.284 4.50e-13 ***
## as.factor(year)2005             75.5298    10.5490   7.160 1.10e-12 ***
## as.factor(year)2006             68.8860    10.5502   6.529 8.18e-11 ***
## as.factor(year)2007             81.1660    10.5502   7.693 2.15e-14 ***
## as.factor(year)2008             93.4260    10.5502   8.855  < 2e-16 ***
## as.factor(year)2009             85.0498    10.5490   8.062 1.22e-15 ***
## as.factor(year)2010             84.1021    10.5538   7.969 2.55e-15 ***
## as.factor(year)2011            102.8260    10.5502   9.746  < 2e-16 ***
## as.factor(year)2012            147.8421    10.5538  14.008  < 2e-16 ***
## as.factor(year)2013            141.9060    10.5502  13.451  < 2e-16 ***
## as.factor(year)2014            162.1829    10.6070  15.290  < 2e-16 ***
## as.factor(year)2015            129.5506    10.5787  12.246  < 2e-16 ***
## as.factor(year)2016            107.1467    10.5917  10.116  < 2e-16 ***
## as.factor(year)2017             94.7821    10.5538   8.981  < 2e-16 ***
## as.factor(year)2018             64.8467    10.5917   6.122 1.09e-09 ***
## as.factor(year)2019            116.9060    10.5502  11.081  < 2e-16 ***
## as.factor(year)2020            133.0860    10.5502  12.615  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 51.3 on 2192 degrees of freedom
## Multiple R-squared:  0.6589, Adjusted R-squared:  0.6438
## F-statistic: 43.66 on 97 and 2192 DF,  p-value: < 2.2e-16
```

```r
summary.regression_2 <- tidy(regression_2, conf.int = T, conf.level = 0.95)

# We are not interested in the constant:
summary.regression_2 <- filter(summary.regression_2, term == "Legalization_medical")

ggplot(summary.regression_2, aes(x = term, y = estimate)) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width=0.2, size=1, color="blue") +
  geom_point(size = 3, color = "red") +
```

```r
geom_hline(yintercept = 0, linetype = 2) +
theme_classic()
```



This time, the effect is negative and not significant anymore. It means that the effect we obtained in the first regression was mostly due to the fact that the number of UFO seeings was more related to particular States rather than to the cannabis legalization.